ANIRUDH BHATTACHARYA

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EDUCATION

University of Southern California, Master of Science – Computer Science — May 2025

University of Mumbai, Bachelor of Technology – Computer Engineering — July 2023

WORK EXPERIENCE

University of Southern California – Marshall School of Business — Los Angeles, CA, USA
Software Engineer — October 2024 – Present
MSME AI Tool | **Tech Stack**: Python, Postgres, React.JS, FastAPI, Jenkins, Docker

- Develop scalable Agentic AI system for MSME marketing, growing operational efficiency by 15% through automated workflows.
- Build fault tolerant, unit/integration tested service to refine decision-making by 30%, automate deployment through CI/CD.

Transmission Line Damage Modeling | **Tech Stack**: Python, ArcGIS, React.JS, Leaflet

- Designed, validated fire-driven transmission line damage models, improving prediction accuracy by 30% over Fragility curves.
- Developed Public Safety Power Shutoff threshold model, reducing asset damage by 15% through cross-functional collaboration.

ViyaMD — Los Angeles, CA, USA
Machine Learning Engineering Intern — May 2024 – July 2024

- Developed communication platforms with Python, Qdrant, Docker connecting doctors, patients, enhancing healthcare delivery.
- Spearheaded construction of internal typing in RAG pipeline improving evaluation metrics, boosting developer efficiency by 5%.
- Optimized ingestion pipeline's support for healthcare guidelines, achieving 90% F1-score, reducing loss for accurate analysis.

University of Southern California – Advanced Composites Simulation Lab — Los Angeles, CA, USA
Machine Learning Engineer, Research — January 2024 – December 2024

- Enhanced aircraft safety by 30% through integrating deep learning models for void detection in aerospace materials (COSB).
- Fine-tuned computer vision models on large-scale 3D micro-CT data, achieving 93% detection score, reducing false negatives.

University of Southern California – Information Technology Program — Los Angeles, CA, USA
Teaching Assistant – ITP 168 — March 2024 – May 2024

- Taught MATLAB to 150+ undergraduates, providing tailored support to boost technical skills and achieve learning goals.
- Designed, evaluated coursework, providing actionable feedback improving student performance, ensure consistent assessment.

KEY PROJECTS

Build-a-Query – pypi.org/project/buildaquery/ — February 2026 – Present

- Implement AST-based Python query builder for 9+ SQL Dialects reducing ORM overhead, enabling flexible OLTP workloads.
- Design compiler with secure parameterization pipeline boosting query safety by 20% by generating injection-safe SQL.

Kernel Craft – pypi.org/project/kernel-craft/ — April 2026 - Present

- Build CUDA/C++ system for sparse conv, fused kernels, ML preprocessing achieving 2x speeds using shared memory upgrades.
- Implement benchmarked GPU ML pipeline cutting memory traffic by 40%, with Python/PyTorch APIs for end-to-end integration.

Offset Store — March 2026 - Present

- Build POSIX shared-memory object store in C eliminating pointer-drift failures in memory by 80% using offset-based addressing.
- Implement sharded shared memory allocator, synchronization enforcing 100% race-safe mutation coverage over access paths.

QuestDB: Automated, Lightweight Snapshots — October 2024 - December 2024

- Reduced time-series database data loss by 50% with automated lightweight snapshots, boosting reliability and data integrity.
- Improved database performance by 40% on unstable hardware through optimizations using Java, Docker, enhancing stability.

Path Planning with Reinforcement Learning — January 2024 - May 2024
Tech Stack: Python, Microsoft AirSim, GCP, OpenAI Gym, OpenCV, PyTorch

- Implemented reinforcement learning training framework enabling real-time pathfinding for UAVs, enhancing mission efficiency.
- Co-developed reward functions based on 3D Image, LIDAR sensors to search 60% faster than standard systems with 0 collisions.

PUBLICATIONS

Feedback Based Telecom Churn Prediction with Machine Learning — doi: 10.1109/ICAST55766.2022.10039530 | IEEE-ICAST 2022

CORE COMPETENCIES AND SKILLS

Languages: Python, Java, JavaScript, C/C++ — **Web:** FastAPI, Django, Flask, React, Node — **Infra**: Docker, Kubernetes, AWS, GCP
AI: PyTorch, Langchain, Tensorflow, SKLearn — **Databases:** Postgres, MongoDB, Redis, SQLite — **Tools:** UNIX, Git, Spark, WSL